

GNR 602 Programming Assignment Topics A.Y. 2025-2026

1. Classify a satellite image using support vector machine classifier with number of classes > 2 , and compare the results of one versus one and one versus rest methods. (LibSVM library or any other suitable library can be used). User specified nonlinear kernel and slack parameters (as supported by the library) may be incorporated. One versus one and one versus rest strategies should be implemented by the student group.
2. Cluster an image using Expectation Maximization (EM) algorithm and comparison with Fuzzy C-Means and evaluate the quality of clustering.
3. Cluster an image using Possibilistic C-Means (PCM) algorithm and compare it with clustering by Kohonen Self-Organizing Map method and evaluate the quality of clustering.
4. Generate Interest Points using SIFT algorithm for airport runways in two high resolution satellite images. Ensure that the runway sizes and orientations are different in the two images. Compare the feature points given by SIFT. The second image can be simulated by rotating and zooming out the first image if suitable image cannot be found from internet.
5. Adapt the Histogram of Oriented Gradients method for extracting all instances of a buildings in the (high resolution) input image and compare it with any given version of YOLO. HOG should be implemented and ready code for YOLO can be used for comparison.
6. Generate NDWI image and make a 3-band image of Green, NIR and NDWI components. Extract all waterbodies from this 3-band image using Mean-Shift segmentation algorithm.
7. Classify a satellite image using a neural network ensemble and majority rule (ensemble is created by varying the number of hidden layers and hidden nodes) and compare the results with classification using a single neural network.
8. Implement a single layer perceptron classifier (input layer + output layer without any hidden layer) with a polynomial function for nonlinear transformation of the input. Compare this result with the result when no nonlinear transformation of the input is done.
9. Extract building outlines from a high resolution satellite image using a combination of Canny edge detector, binary morphological operations for clean-up of the edges, and Hough transform. Hint: Suppress non-built up areas in the image using Normalized Difference Building Index (NDBI).
10. Smooth an image using Daubechies wavelet (db-4 or db-6) and segment it using quadtree based region segmentation method. Compare the result with any other method (readymade code allowed for comparison).
11. Use ant colony optimization approach and reduce dimensionality of a hyperspectral image. Compare the classification accuracy obtained using the reduced dimensional data with accuracy obtained using the full dimensional data.
12. Sharpen a multispectral image using wavelet transform and PCA and compare the result with wavelet transform and original image.
13. Initialize a neural network with genetic algorithm given weights and classify a satellite image. Compare the results with a neural network initialized by random weights.
14. Extract endmembers from an MNF based dimensionality reduced hyperspectral image. Use Vertex Component Analysis approach and compare the result with PPI given endmembers. (PPI result can be obtained using any standard software package or readymade code).
15. Make a collage of Brodatz textures as the test image. Reduce number of gray levels in the image by applying multi-threshold version of Otsu method (readymade code allowed for this step). Compute singular values of the reduced gray level co-occurrence matrix and selected Haralick texture features from the same reduced gray level co-occurrence matrix (Co-occurrence matrix and Haralick texture features are to be implemented by the group, singular value computation can be done using readymade code or library function). Segment the collage image based on the two feature sets using K-Means algorithm and compare the results.
16. Extract building corners from a high resolution satellite image using the original and scale invariant versions of Harris corner detector and compare results. Simulate a lower-resolution version of the high resolution image for comparison purpose.